

11. Sequences and Series

11.2. Sequences

11.1b Learning Objectives

Theorem 11.2.1 Divergence Theorem (Test for Divergence)

If the series $\sum_{n=1}^{\infty}$ is convergent, then $\lim_{n \rightarrow \infty} a_n = 0$.

Therefore,

- If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent.
- If $\lim_{n \rightarrow \infty} a_n = 0$, then the series may converge or diverge. The test is inconclusive.

Example 11.2.1

Use the divergence test to determine the divergence of the following series if applicable.

$$\sum_{k=1}^{\infty} \frac{k^2}{k^2 - 2k + 5}$$

Solution

Looking at the limit of the terms we have

$$\lim_{k \rightarrow \infty} \frac{k^2}{k^2 - 2k + 5} \stackrel{\text{LH}}{=} \lim_{k \rightarrow \infty} \frac{2k}{2k - 2} \stackrel{\text{LH}}{=} \lim_{k \rightarrow \infty} \frac{2}{2} = 1 \neq 0$$

Thus, by the Divergence Theorem, the series diverges.

Example 11.2.2

Use the divergence test to determine the divergence of the following series if applicable.

$$\sum_{n=1}^{\infty} \ln \left(\frac{n^2 + 1}{2n^2 + 1} \right)$$

Solution

Looking at the limit of the terms we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \ln \left(\frac{n^2 + 1}{2n^2 + 1} \right) &= \ln \left(\lim_{n \rightarrow \infty} \frac{n^2 + 1}{2n^2 + 1} \right) \\ &\stackrel{\text{LH}}{=} \ln \left(\lim_{n \rightarrow \infty} \frac{2n}{4n} \right) \\ &= \ln \left(\frac{1}{2} \right) \neq 0 \end{aligned}$$

Thus, by the Divergence Theorem, the series diverges.

11.1b Section Summary: